

Researcher: Vanessa Kountakis

Research Question: Do Transportation Habits Predict Communication Apprehension?

Answer, Response, + Summary of Results

A one-way ANOVA found no significant differences in communication apprehension across public transportation frequency groups, $F(4, 255) = 2.36, p = 0.054$. In contrast, communication apprehension differed significantly across rideshare frequency groups, $F(4, 255) = 9.39, p < .001$. Tukey comparisons showed that participants who used rideshare 2–4 days per month, 4–8 days per month, or 8 or more days per month had significantly lower communication apprehension than participants who never used rideshare.

Supplemental Pearson correlations showed a small negative relationship between public transportation frequency and communication apprehension, $r = -0.161, p = 0.009$, and a stronger negative relationship between rideshare frequency and communication apprehension, $r = -0.344, p < .001$.

Overall, rideshare frequency showed the clearest relationship with communication apprehension, with more frequent rideshare users generally reporting lower communication apprehension.

What Questions or Uncertainties Remain?

Would rideshare frequency remain associated with communication apprehension after accounting for factors such as age, car access, and employment status?

Appendix

Figure 1: Transportation Habits and Communication Apprehension

```
ggplot(  
  transport_plot,  
  aes(  
    x = frequency,  
    y = ca_score  
  )  
) +  
  geom_boxplot() +  
  facet_wrap(  
    ~ transport_type,  
    ncol = 2  
  ) +  
  labs(  
    x = "Frequency of Use",  
    y = "Communication Apprehension Score"  
  ) +  
  theme_minimal(base_size = 9) +  
  theme(  
    axis.text.x = element_text(  
      angle = 25,  
      hjust = 1  
    )  
  )  
)
```

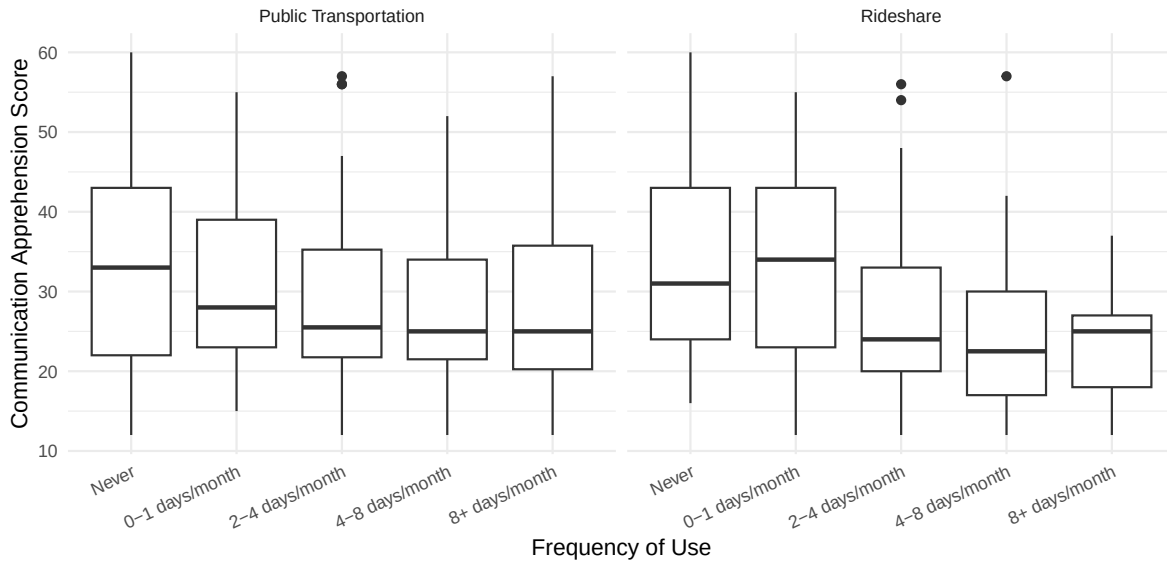


Figure 1: Communication apprehension scores across public transportation and rideshare frequency groups.

Table 1: ANOVA Results

```
anova_table |>
  mutate(
    sumsq = round(sumsq, 2),
    meansq = round(meansq, 2),
    statistic = round(statistic, 2),
    p.value = case_when(
      is.na(p.value) ~ "",
      p.value < .001 ~ "< .001",
      TRUE ~ sprintf("%.3f", p.value)
    )
  ) |>
  kable(
    col.names = c(
      "Analysis",
      "Source",
      "df",
      "Sum of Squares",
      "Mean Square",
      "F",
      "p"
    ),
  ),
```

```
caption = "One-way ANOVA results for transportation frequency and communication apprehension."
)
```

Table 1: One-way ANOVA results for transportation frequency and communication apprehension.

Analysis	Source	df	Sum of Squares	Mean Square	F	p
Public Transportation	Between Groups	4	1222.84	305.71	2.36	0.054
Public Transportation	Residuals	255	33070.63	129.69	NA	
Rideshare	Between Groups	4	4401.41	1100.35	9.39	< .001
Rideshare	Residuals	255	29892.06	117.22	NA	

Table 2: Supplemental Pearson Correlations

```
correlation_table |>
  mutate(
    r = round(r, 3),
    p = case_when(
      p < .001 ~ "< .001",
      TRUE ~ sprintf("%.3f", p)
    )
  ) |>
  kable(
    col.names = c(
      "Transportation Measure",
      "r",
      "p"
    ),
    caption = "Supplemental Pearson correlations between transportation frequency and communication a
  )
```

Table 2: Supplemental Pearson correlations between transportation frequency and communication apprehension.

Transportation Measure	r	p
Public Transportation	-0.161	0.009
Rideshare	-0.344	< .001

Note. Transportation frequency was coded from 0 = Never to 4 = 8 or more days per month for the supplemental Pearson correlations.